

```
In[*]:= (*determinación de Q*)
datafreond1 = {{0, Log[1.151067715]}, {1.0848825, Log[1.129547402]},
               |logaritmo |logaritmo
               {2.169765, Log[1.125339542]}, {3.2546475, Log[1.11882716]},
               |logaritmo |logaritmo
               {4.33953, Log[1.110005359]}, {5.4244125, Log[1.102326289]},
               |logaritmo |logaritmo
               {6.509295, Log[1.094670089]}, {7.5941775, Log[1.087608761]},
               |logaritmo |logaritmo
               {8.67906, Log[1.079833184]}, {9.7639425, Log[1.072564539]},
               |logaritmo |logaritmo
               {10.848825, Log[1.065314819]}, {11.9337075, Log[1.058316911]},
               |logaritmo |logaritmo
               {13.01859, Log[1.050876939]}, {14.1034725, Log[1.044442844]},
               |logaritmo |logaritmo
               {15.188355, Log[1.037121808]}, {16.2732375, Log[1.03070799]}};
```

```
In[*]:= datafit1 = NonlinearModelFit[datafreond1, {t * a + b}, {a, b}, t]
               |ajusta a modelo no lineal
```

Out[*]=

```
FittedModel[0.133083 - 0.00640336 t]
```

```
In[*]:= datafit1[{"BestFit", "ParameterTable"}]
```

Out[*]=

		Estimate	Standard Error	t-Statistic	P-Value
{0.133083 - 0.00640336 t,	a	-0.00640336	0.000122613	-52.2244	1.90199 × 10 ⁻¹⁷
	b	0.133083	0.00117103	113.646	3.65729 × 10 ⁻²²

```
In[*]:= pplot1 = ListPlot[datafreond1,
               |representación de lista
```

```
PlotStyle → {Black, Thickness[0.003]}, PlotLegends → {"Datos experimentales"}];
               |estilo de repre... |negro |grosor |leyendas de representación
```

graph1 =

```
Plot[datafit1[t], {t, -0.1, 17}, BaseStyle → {FontFamily → "Calibri", FontSize → 14},
               |representación gráfica |estilo base |familia de tipo de letra |tamaño de tipo de letra
```

```
PlotStyle → {Blend[{Blue, White}], Thick},
               |estilo de repre... |mezcla... |azul |blanco |grueso
```

```
PlotRange → {{-0.1, 17}, {0.02, 0.15}}, Frame → True,
               |rango de representación |marco |verdadero
```

```
FrameLabel → {Style["t / s", Bold, 16], Style["ln(v / m s-1", 16, Bold]}];
               |etiqueta de marco |estilo |negrita |estilo |negrita
```

```
Needs["ErrorBarPlots`"];
               |necesita
```

... General: ErrorBarPlots` is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

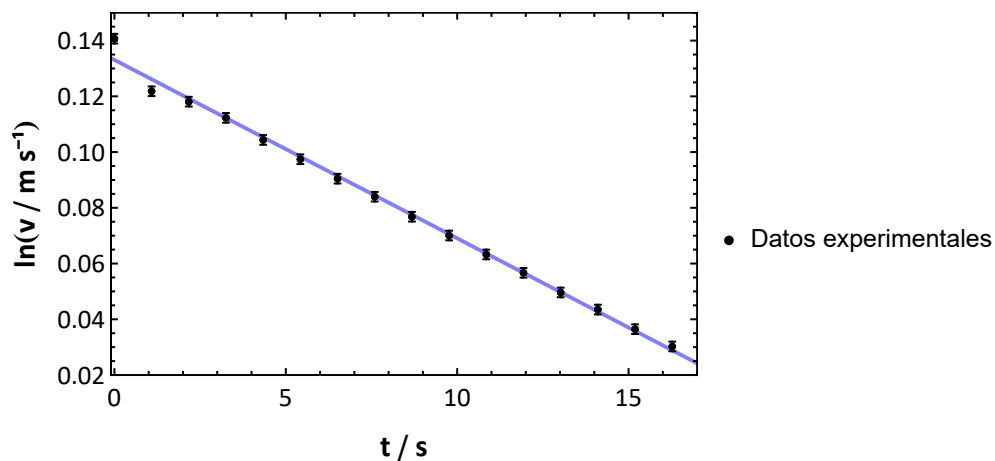
```

In[*]:= dataerr1 = ErrorListPlot[{{0, Log[1.151067715]}, ErrorBar[0, 0.001724595]},
                                |logaritmo
                                {{1.0848825, Log[1.129547402]}, ErrorBar[3.96246 * 10^(-5), 0.001724578]},
                                |logaritmo
                                {{2.169765, Log[1.125339542]}, ErrorBar[7.92493 * 10^(-5), 0.001724575]},
                                |logaritmo
                                {{3.2546475, Log[1.11882716]}, ErrorBar[0.000118874, 0.00172457]},
                                |logaritmo
                                {{4.33953, Log[1.110005359]}, ErrorBar[0.000158499, 0.001724563]},
                                |logaritmo
                                {{5.4244125, Log[1.102326289]}, ErrorBar[0.000198123, 0.001724557]},
                                |logaritmo
                                {{6.509295, Log[1.094670089]}, ErrorBar[0.000237748, 0.001724551]},
                                |logaritmo
                                {{7.5941775, Log[1.087608761]}, ErrorBar[0.000277372, 0.001724546]},
                                |logaritmo
                                {{8.67906, Log[1.079833184]}, ErrorBar[0.000316997, 0.00172454]},
                                |logaritmo
                                {{9.7639425, Log[1.072564539]}, ErrorBar[0.000356622, 0.001724535]},
                                |logaritmo
                                {{10.848825, Log[1.065314819]}, ErrorBar[0.000396246, 0.001724529]},
                                |logaritmo
                                {{11.9337075, Log[1.058316911]}, ErrorBar[0.000435871, 0.001724524]},
                                |logaritmo
                                {{13.01859, Log[1.050876939]}, ErrorBar[0.000475496, 0.001724519]},
                                |logaritmo
                                {{14.1034725, Log[1.044442844]}, ErrorBar[0.00051512, 0.001724514]},
                                |logaritmo
                                {{15.188355, Log[1.037121808]}, ErrorBar[0.000554745, 0.001724509]},
                                |logaritmo
                                {{16.2732375, Log[1.03070799]}, ErrorBar[0.000594369, 0.001724504]}},
                                AxesOrigin -> {0, 0}, PlotStyle -> {Black, Thickness[0.003]};
                                |origen de ejes |estilo de repre... |negro |grosor

In[*]:= Show[graph1, pplot1, dataerr1]
                                |muestra

```

Out[*]=



```
In[*]:= (*determinación de I*)
```

[número i]

```
datafreond2 = {{9.80665, 29.8116},
  {9.769332736, 29.9209}, {9.657664951, 29.5936}, {9.472496504, 29.0521},
  {9.21523664, 28.1961}, {8.88784326, 27.1441}, {8.492808026, 25.6036},
  {8.033137395, 24.1081}, {7.512329738, 22.5625}, {6.934348716, 20.7025},
  {6.303593113, 18.5761}, {5.62486336, 16.5649}, {4.903325, 14.2884}};
```

```
datafit2 = NonlinearModelFit[datafreond2, {g * a + b}, {a, b}, g]
```

[ajusta a modelo no lineal]

```
Out[*]=
```

```
FittedModel[[-1.6403 + 3.22566 g]]
```

```
In[*]:= datafit2[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

		Estimate	Standard Error	t-Statistic	P-Value
{ -1.6403 + 3.22566 g,	a	3.22566	0.0213877	150.819	1.36448×10^{-19}
	b	-1.6403	0.175479	-9.34756	1.44464×10^{-6}

```
In[*]:= pplot2 = ListPlot[datafreond2,
```

[representación de lista]

```
PlotStyle -> {Black, Thickness[0.003]}, PlotLegends -> {"Datos experimentales"}];
```

[estilo de repre... [negro [grosor

[leyendas de representación]

```
graph2 =
```

```
Plot[datafit2[g], {g, 4.5, 10}, BaseStyle -> {FontFamily -> "Calibri", FontSize -> 14},
```

[representación gráfica

[estilo base

[familia de tipo de letra

[tamaño de tipo de leti

```
PlotStyle -> {Blend[{Blue, White}], Thick},
```

[estilo de repre... [mezcla... [azul [blanco [grueso

```
PlotRange -> {{4.7, 10}, {13, 31}}, Frame -> True,
```

[rango de representación

[marco

[verdadero

```
FrameLabel -> {Style["g_e / m s^-2", Bold, 16], Style["\omega_o^2 / rad^2 s^-2", 16, Bold]}];
```

[etiqueta de marco [estilo

[negrita

[estilo

[negrita

```
Needs["ErrorBarPlots`"];
```

[necesita

```
In[*]:= dataerr2 = ErrorListPlot[{{9.80665, 29.8116}, ErrorBar[0, 0.024024]},
```

```
{9.769332736, 29.9209}, ErrorBar[0.001491743, 0.01094]},
```

```
{9.657664951, 29.5936}, ErrorBar[0.002972133, 0.0022848]},
```

```
{9.472496504, 29.0521}, ErrorBar[0.004429904, 0.0023716]},
```

```
{9.21523664, 28.1961}, ErrorBar[0.00585396, 0.0070092]},
```

```
{8.88784326, 27.1441}, ErrorBar[0.007233464, 0.004689]},
```

```
{8.492808026, 25.6036}, ErrorBar[0.008557917, 0.0037444]},
```

```
{8.033137395, 24.1081}, ErrorBar[0.009817239, 0.0033388]},
```

```
{7.512329738, 22.5625}, ErrorBar[0.011001845, 0.00304]},
```

```
{6.934348716, 20.7025}, ErrorBar[0.012102722, 0.002821]},
```

```
{6.303593113, 18.5761}, ErrorBar[0.013111489, 0.0031032]},
```

```
{5.62486336, 16.5649}, ErrorBar[0.01402047, 0.0027676]},
```

```
{4.903325, 14.2884}, ErrorBar[0.014822746, 0.0031752]}] AxesOrigin ->
```

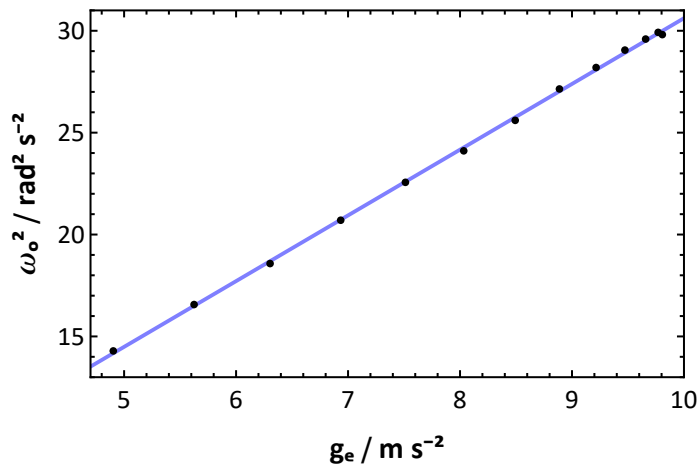
[origen de ejes

```
{0, 0}, PlotStyle -> {Black, Thickness[0.003]}];
```

[estilo de repre... [negro [grosor

```
In[*]:= Show[graph2, pplot2, dataerr2]
      muestra
```

```
Out[*]=
```



• Datos experimentales

```
In[*]:= (*Cálculo v0 y tau*)
```

```
In[*]:= v0 = E ^ (0.13308348474422863)
      número e
```

```
Out[*]=
```

1.14235

```
In[*]:= errrv0 = v0 * 0.0011710317271807547
```

```
Out[*]=
```

0.00133772

```
In[*]:= τ = 1 / (2 * 0.006403363928602255)
```

```
Out[*]=
```

78.084

```
In[*]:= errτ = 0.00012261260140726573 / (2 * 0.006403363928602255 ^ 2)
```

```
Out[*]=
```

1.49516

```
In[*]:= Q[T_, τ_] := ((2 * Pi) / T) * τ;
      número pi
```

```
In[*]:= Q[2.16977, 78.0839579906778]
```

```
Out[*]=
```

226.114

```
In[*]:= errQ = Sqrt[ ((2 * Pi / 2.16977) * 1.495163686487911) ^ 2 + ((2 * Pi / 2.16977 ^ 2) * 0.00004) ^ 2]
      raíz cuadrada número pi      número pi
```

```
Out[*]=
```

4.32967